

Exam Date & Time: 25-Jun-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Makeup Examination - June -2024
Bio-Informatics [CSE-4012]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) With a schematic diagram discuss the constituent components of the UniProt Database and also elaborate on the six steps involved in the UniProtKB manual curation process in protein synthesis in reference to Swiss-Prot. (4)
 - A)
 - B) Distinguish between primary databases and derivative databases in the context of Biological databases by giving at least two examples for each. Discuss any four features of the RefSeq Nucleotide Databases (4)
 - C) Determine the highest matched score for aligning AGTCTAG with AGATTCGAG by keeping the mismatches and gaps to the minimum using a simple nucleotide alignment strategy. (2)
- 2) Justify with at least three valid reasons for the need to align biological sequences pertaining to genetic nucleotides and proteins. Discuss at least three applications where the alignment of genetic sequences has proven beneficial to humankind and society at large. (4)
 - A)
 - B) Interpret the significance and working of ProDom, SMART, RPS-BLAST, and CDART in the context of integration of biological databases, statistical models, and other tools for multiple sequence alignment. (4)
 - C) Illustrate the functionalities of PROSITE and EMotif in carrying out multiple sequence alignment. (2)
- 3) Discuss Block-Based Alignment by citing DIALIGN2 and Match-Box. Highlighting the need for Motifs for Protein Alignment, interpret how Regular Expressions are required for the correlation of Motifs and Domain Databases during synthesis of proteins. (4)
 - A)
 - B) "Molecular Phylogenetics involves the derivation of Phylogenetic tree at a Genetic Level". Discuss any two Clustering-Based methods for the construction of a Phylogenetic tree. (4)
 - C) Interpret the primary objective of the GOR Algorithm in the context of secondary structure prediction. (2)
- 4) With an illustration using a flow diagram, discuss the steps in detail, the Homology (5)

Based Comparative Modelling algorithm. Give at least two differences between a Phylogram and a Cladogram.

A)

B) Point out the integration of Pharmacology and Bioinformatics and bring out their applications. Discuss at least three major medical applications of Bioinformatics in detail. (5)

5) Discuss in detail the steps to predict the tertiary structure of a peptide using homology modelling approach with SWISS-MODEL online tool. (5)

A)

B) Give the formal definition of an Ontology and a Taxonomy and list at least two differences between them. Discuss the constituents of EcoCyc Ontology, the Gene Ontology, and the TAMBIS Ontology in Bioinformatics. (5)

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